Evaluating Football Player 1v1 Ability Using Elo Rating Models

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Abstract

Football clubs increasingly rely on data analytics for scouting, opposition analysis, and performance review. Despite this, many of the indicators used for this work don't take into account the quality of opposition a player actually faced. This gap shows up most clearly in metrics for 1v1 duels and dribbles, which record only success or failure and treat every opponent as interchangeable; a player who repeatedly targets weaker defenders can end up with a better record than one who takes on stronger opposition more often, regardless of who is the better dribbler. This paper builds separate attacking and defensive Elo ratings from StatsBomb's open event data for the 2015/16 season and compares them against traditional success-rate metrics. Data comes from three leagues, the Premier League, La Liga, and Serie A (the Bundesliga and Ligue 1 were excluded due to their 2015/16 matches not being fully available through open access), covering 1,140 matches and 37,759 attacker-defender duel pairs. Elo ratings (base 1,000, $K = 32$) were calculated independently within each league, then pooled between leagues to test whether the resulting patterns hold statistically in aggregate. In the pooled sample, Elo tracked success rate strongly (attacking $r = 0.84$, defending $r = 0.90$, both $p < .0001$), a relationship that held individually across all three leagues ($r = 0.82$–0.86 attacking, 0.89–0.92 defending). Within the Premier League, individual players diverged sharply between the two rankings: Jamie Vardy, for example, ranked 29th by Elo but 227th by success rate among 297 qualifying attackers. Attacking and defensive Elo were only weakly related within players (pooled $r = 0.076$), and leagues differed measurably in defensive difficulty, from a mean defensive success rate of 0.383 in the Premier League to 0.430 in Seria A. This all gives the result that Elo offers a statistically robust, opposition-adjusted complement to success rate within a given league.

1 Introduction

1.1 The role of data analytics in football

Data analytics is key to the everyday operations of professional football clubs, feeding into recruitment, opposition scouting, and post-match review. Recent growth in football analytics can be broken down into three main trends. The first of these is the development of evaluation models such as expected goals and pitch control that give coaches interpretable and actionable metrics (Pu et al., 2024). The second is the growing accessibility of specialized tracking and event data from commercial vendors (e.g., Opta, StatsBomb, SkillCorner, Second Spectrum) (Pu et al., 2024). Finally, the use of computer vision and statistical learning methods within soccer has meant that tasks such as event annotation, once done manually, are increasingly handled through automated detection systems (Pu et al., 2024).

1.2 Origins of Elo-style rating systems

The invention of Elo rating models can be traced all the way back to the 1950s for it's use in competitive chess rankings (Glickman, n.d.). In chess these ratings have various uses, including, the seeding of tournaments, use as barriers of entry for high level tournaments, or most commonly as a method of tracking a competitors progress, skill level, or equivalent competition (Glickman, n.d.). The basic chess Elo model has evolved since it's original conception to become what we know it as today. Glickman breaks down this current model when saying:

> *The Elo rating system assigns to every player a numerical rating based on performances in competitive chess. A rating is a number normally between 0 and 3000 that changes over time depending strictly on the outcomes of tournament games. When two players compete, the rating system predicts that the one with the higher rating is expected to win more often than the lower rated player. The more marked the difference in ratings, the greater the likelihood that the higher rated player will win.*

(Glickman, n.d.)

1.3 Elo ratings across sport

The same model has since spread well beyond chess. Some examples include Men's professional tennis, which has incorporated rating-style adjustments into elements of its ranking system (ATP Tour, 2024), and FIFA which have used an Elo-based formula for its men's world rankings since 2018 (FIFA, 2024). David Aldous argues that Elo-type models suit a wide range of sports precisely because they require only a sequence of pairwise results rather than detailed performance data, which is one reason the approach generalises so readily across disciplines (Aldous, 2017). Despite this lack of data requirements, Aldous still identifies that Elo models are widely unexplored and neglected in current sports analysis (Aldous, 2017).

1.4 Elo ratings in football

Within football specifically, published applications of Elo-style ratings have mostly stayed at the level of whole teams, used to rank or compare squads rather than individual players. Kim and Kim note that player-level applications, and in particular ratings for a specific in-game skill such as the 1v1 duel, are largely absent from the football analytics literature, despite duels being a common and clearly defined event type in modern tracking and event data (Kim & Kim, 2025).

1.5 Limitations of raw success rate

Traditional 1v1 metrics report only a raw success rate: the proportion of duels or dribble attempts a player wins. McDermott, Collins, Mangan, and Warne point out that a success rate calculated this way cannot separate a player who wins duels against weak opposition from one who wins the same proportion against elite defenders, because the metric carries no information about who was actually faced (McDermott, Collins, Mangan, & Warne, 2021). This is the specific gap the present study addresses.

1.6 Aim and objectives

This study aims to construct separate attacking and defensive Elo ratings from StatsBomb's open event data within each of the three selected major European leagues, and to test whether these ratings produce player rankings that differ meaningfully from those produced by raw success rate. A second aim is to pool the results across the three leagues to test whether the relationship between Elo and success rate, and the positional patterns found within each league, are statistically robust once the sample size is maximised. Despite this combination of leagues for positional analysis, still note that this doesn't imply relativity between ratings in different leagues.

Three main objectives follow from these aims. The first is to extract and define 1v1 dribble duel pairings separately within each of the three leagues. The second is to construct attacking and defensive Elo ratings for each league independently. The third is to compare Elo-derived rankings against success-rate rankings within each league, while separately checking using the pooled sample, whether the strength and direction of these effects (correlation size, positional ANOVA effect sizes) are consistent across leagues.

Note that the Bundesliga and Ligue 1 were excluded from all three objectives. This reflects a constraint on data access rather than a deliberate design choice: not every Bundesliga or Ligue 1 match from the 2015/16 season was available through StatsBomb's free open data at the time this study was carried out.

2 Methodology

2.1 Data source

Event data for this study came from StatsBomb's open data, accessed through the `statsbombpy` package (StatsBomb, 2026). Three competitions from the 2015/16 season were used: the Premier League (380 matches), La Liga (380 matches), and Serie A (380 matches) giving 1,140 matches in total. The Bundesliga and Ligue 1 were excluded because not all of their matches from that season were available through open access.

2.2 Defining 1v1 duel events

The used definition for duel events for this paper was defined as any StatsBomb `Dribble` event, with the player recorded on that event treated as the attacker and the outcome taken from the `dribble_outcome` field (*Complete* or otherwise). Each dribble was linked to its opposing defender using StatsBomb's `related_events` field, which cross-references connected events within the same passage of play. For dribbles recorded as *Complete*, the defender was taken to be the player logged on the corresponding `Dribbled Past` event. For dribbles recorded as anything other than *Complete*, the defender was taken to be the player logged on the corresponding `Duel` event, restricted to duels where `duel_type` was *Tackle*. Where an unsuccessful dribble had no resolvable link via `related_events`, it was instead paired with a tackle-type `Duel` event sharing the same match, period, and timestamp, as a fallback. Pairings in which the attacker and defender resolved to the same player, or for which no defender could be identified by either method, were discarded, as were duplicate pairings arising from repeated links between the same match, timestamp, attacker, and defender. Each retained pairing was coded as a binary outcome (1 = attacker win, 0 = defender win), corresponding to a completed and an incompleted dribble respectively.

Table 1 reports the number of qualifying attacker-defender pairs identified within each league, together with the resulting attacker win rate. The pooled row is a total of the three independent per-league totals, included to describe the overall scale of the dataset. This does not imply that an attacker from one league ever faced a defender from another.

Table 1: Attacker–defender duel pairs by league.

League	**Pairs**	**Attacker wins**	**Defender wins**	**Attacker win %**
Premier League	12,983	8,050	4,933	62.0%
La Liga	13,707	8,026	5,681	58.6%
Serie A	11,069	6,368	4,701	57.5%
Pooled (aggregate only)	**37,759**	**22,444**	**15,315**	**59.4%**

2.3 Minimum duel threshold

While sample size is important to this study, a minimum requirement of 10 duels within a league was required for a player to be included in the overall Elo rankings and analysis. This threshold was included to reduce the influence of unstable ratings within the first few encounters for a player, as ratings based on only a few duels can only be considered provisional and lack proper data to stabilize (Glickman & Jones, 1999).

Elo ratings were computed independently within each league. A player's numeric rating is only meaningful relative to the opponents they actually faced inside their own league's duel network. Pooling player-level rows across leagues for the descriptive, correlation, and ANOVA statistics that follow does not mean these numeric ratings are directly comparable to one another in absolute terms, since there is no shared duel network across leagues against which such a comparison could be calibrated. Future research using multiple seasons or inter-league competitions such as the UEFA Champions League, may allow for these results to have substantial meaning across leagues.

Table 2: Unique rated players and qualifying samples (≥10 duels) by league.

League	**Unique players rated**	**Qualifying attackers**	**Qualifying defenders**
Premier League	509	297	336
La Liga	506	316	355
Serie A	503	287	336
Pooled total (aggregate only)	**1,504**	**899**	**1,026**

2.4 Position-group mapping

Position groups were applied for the purpose of viewing trends within and between different position groups. Players primary positions were applied using the mode of all their primary positions across all duel events in which they participated. To get their position for a specific duel event, the StatsBomb field `position` was used. This method was preferred over using a current squad-list label, as a squad-list label doesn't hold up well over a season. This is mainly due to the fact that players get moved around through rotation, injury cover, or in-game switches, so the modal label gives a more honest picture of where someone actually played. For the purpose of readability, StatsBomb's granular position labels were grouped together into 8 main position groups, combining labels such as *Left Centre Back* and *Right Centre Back* into one group. These final groups were: Centre backs, Full backs, Wing backs, Defensive mids, Centre mids, Attacking mids, Wingers, and Strikers (Note Table 11 for complete breakdown).

2.5 Elo model specification and analysis plan

For each duel pairing, the model produces an expected outcome from the two players' current ratings before comparing it against what actually happened. The standard logistic form Elo from chess was used (Glickman, n.d.):

$$E_a = \frac{1}{1 + 10^{(R_d - R_a)/400}}$$

where R_a is the attacker's rating and R_d is the defender's rating. The defender's expected score follows directly from this as$E_d = 1 - E_a$ (Glickman, n.d.). For more detail on this win probability, read Bisberg and Cardona-Rivera (2019), as it's methods were followed directly.

For updating the players rating after a duel resolves, both ratings move by the gap between what happened and what was expected, scaled by a constant K (K is a scaling factor that is used to define how volatile a single duel is to both players individual ratings). For updating these ratings the following formula is used:

$$R_a \leftarrow R_a + K(S_a - E_a), \qquad R_d \leftarrow R_d + K(S_d - E_d)$$

where $S_a \in \{0, 1\}$ marks is the value of whether the attacker won the duel and $S_d = 1 - S_a$ is for the defender (1 if they actually won and 0 if not) (Glickman, n.d.). Updates were applied duel by duel, in chronological order, rather than aggregated to the match or season level, so a player's rating at any point reflects everything they had done up to that moment in the data. This does mean that if duels occured in different orders, ratings could change.

Each player carries two ratings, one for attacking duels and one for defending them, and only the rating for the role a player occupied in a given duel is updated. Bisberg and Cardona-Rivera (2019) point out that most Elo adaptations have some sport-specific correction applied onto the baseline model, whether this be, home advantage, margin of victory, and so on, to fit the shape of the competition being modeled. Splitting attack from defense is the sport-specific adjustment made here. This originates from the fact that duel is not a symmetric event, winning it as an attacker and winning it as a defender draw on different qualities, and folding both into one blended rating would erase that distinction rather than capture it. In further studies, one may decide to apply multiple other corrections on to this model, for example adding in home advantage.

For initialization, all players start at a rating of 1,000. Elo himself noted that a rating scale has no natural fixed point, and that a new pool of competitors has to be seeded somewhere before ratings can differentiate through play (Glickman, n.d.). For this a round number is picked rather than inferring a starting rating from limited prior information, since nothing in the data gives a set way to do so.

K was fixed at 32 and applied uniformly across all duels, with no margin-of-victory term (often applied to team Elo ratings to reflect that the margin of a win should make the rating change more volatile). Bisberg and Cardona-Rivera (2019) grid-searched K for a comparable Elo model built on esports data and found accuracy was maximized at a fairly low value ($K = 5$), while log loss favoured the opposite end of their tested range ($K = 50$). A bare-minimum model with $K = 1$ and no further adjustments only reached 56% accuracy. Football and basketball adaptations of Elo, by contrast, have typically used K somewhere between 30 and 50. Thus a K value of 32 was selected given the recommended K values for similar sports analyzed by Bisberg and Cardona-Rivera (2019).

Alongside the Elo ratings, a plain success rate was recorded for every player in both roles, duels won divided by duels played. This sits outside the Elo model itself, and only exists to provide a baseline against which Elo is compared for the rest of the paper. The whole point of this comparison is to see where the two rankings pull apart. A player who wins a high share of duels against weak opposition should, in principle, look better on success rate than on Elo. It is exactly that gap created by opposition strength looks to explore.

3 Findings

3.1 Descriptive statistics

Table 3 and Table 4 report descriptive statistics for attacking and defensive Elo and success rate, by league and pooled.

Table 3: Attacking Elo / success rate, by league and pooled (aggregate statistic only).

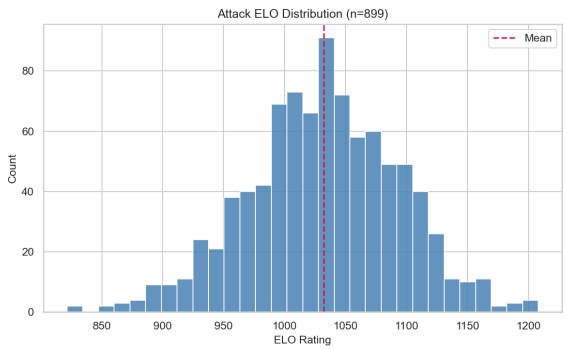
League	n	Elo mean (SD)	Elo skew/kurt	SR mean (SD)
Premier League	297	1041.03 (62.12)	−0.065 / −0.235	0.628 (0.125)
La Liga	316	1031.18 (63.28)	−0.298 / 0.357	0.597 (0.133)
Serie A	287	1023.63 (58.91)	−0.109 / 0.132	0.580 (0.125)
Pooled	**899**	**1032.08 (62.01)**	−.148 / 0.092	**0.602 (0.129)**

Table 4: Defensive Elo / success rate, by league and pooled (aggregate statistic only).

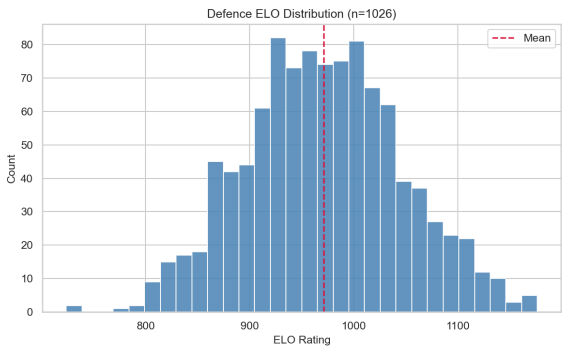
League	n	Elo mean (SD)	Elo skew/kurt	SR mean (SD)
Premier League	336	963.75 (75.04)	0.073 / 0.132	0.383 (0.144)
La Liga	355	971.00 (78.21)	0.035 / −0.508	0.408 (0.154)
Serie A	336	977.73 (71.02)	0.091 / −0.342	0.430 (0.150)
Pooled	**1,026**	**970.82 (75.13)**	0.046 / −0.244	**0.407 (0.150)**

Across leagues, defensive Elo and success rate move together. The Premier League has the lowest mean defensive Elo and success rate of the three leagues, and Serie A the highest. Attacking Elo shows a milder pattern running the other way, highest in the Premier League and lowest in Serie A. Every Elo distribution, in each league and pooled, stays close to symmetric, with skew ranging from −0.30 to +0.09, which supports the use of Pearson correlation throughout the analyses that follows.

3.2 Elo rating distributions



(a) Attacking Elo (pooled).



(b) Defensive Elo (pooled).

Figure 1: Pooled attacking and defensive Elo rating distributions across the three leagues. Individual-league distributions are reported separately in Appendix C.

Figure 1 supports the symmetry claim made in Section 3.1. Both pooled distributions are close to bell-shaped, consistent with the low skew values reported in Tables 3 and 4. Two further features are visible only at league level (Appendix C, Figures 5 and 3), these being that Serie A shows a thin right tail on attacking Elo, with few players reaching the higher ratings, while the Premier League shows a thin left tail on the same measure, with few players falling to the lowest ratings.

3.3 Correlation between Elo and success rate

Table 5 reports the correlation between Elo and success rate, by league and pooled; all correlations are significant at $p < .0001$.

Table 5: Correlation of Elo with success rate, by league and pooled (all $p < .0001$).

League	Attack Pearson r	Attack Spearman r	Defense Pearson r	Defense Spearman r
Premier League	0.8168	0.8261	0.8915	0.8989
La Liga	0.8499	0.8833	0.8977	0.9284
Serie A	0.8570	0.8725	0.9122	0.9323
Pooled	**0.8430**	**0.8640**	**0.8999**	**0.9213**

The pooled correlation is the headline statistical result. Both attacking Elo ($r = 0.843$) and defensive Elo ($r = 0.899$) track success rate strongly. Within the three individual leagues, the Premier League shows the weakest relationship of the three in both roles and Serie A the strongest. Despite this all leagues still have significant correlation confirming that the pooled figure isn't hiding a league where the relationship breaks down.

3.4 ANOVA by position group

Table 6 and Table 7 report one-way ANOVA results for Elo and success rate by position group, by league and pooled; both roles are significant at $p < .0001$ throughout.

Table 6: Attacking Elo / success rate by position group — ANOVA by league and pooled.

League	Elo F	Elo η^2	SR F	SR η^2
Premier League	7.4289	0.133	19.2123	0.284
La Liga	5.4818	0.096	16.9630	0.248
Serie A	5.4689	0.113	13.4170	0.239
Pooled	**13.0755**	**0.093**	**38.9336**	**0.234**

Table 7: Defensive Elo / success rate by position group — ANOVA by league and pooled.

League	Elo F	Elo η^2	SR F	SR η^2
Premier League	33.9027	0.382	53.5650	0.494
La Liga	43.4583	0.428	62.5555	0.519
Serie A	37.2112	0.430	40.4767	0.451
Pooled	**94.1114**	**0.393**	**128.1203**	**0.468**

The pooled attacking-Elo effect size ($\eta^2 = 0.093$) is smaller than every individual league's own effect size (0.096–0.133). Section 4.3 discusses this further, and what it says about the effect of the pooled sample vs. individual leagues. Alternatively the pooled defensive-Elo effect size ($\eta^2 = 0.393$),sits comfortably inside the individual-league range (0.382–0.430), with no equivalent drop.

3.5 Attack vs defence Elo correlation (within-player)

Table 8 reports the correlation for individual players between attacking and defensive Elo, by league and pooled.

Table 8: Within-player attack–defence Elo correlation, by league and pooled.

League	**Pearson** r	p	n
Premier League	0.0252	0.685 (n.s.)	262
La Liga	0.1328	0.026 (sig.)	281
Serie A	0.0924	0.147 (n.s.)	247
Pooled	**0.0757**	**0.033 (sig.)**	**790**

The relationship is weak in every league ($r < 0.15$ throughout) and inconsistent in statistical significance, reaching significance in La Liga but not in the Premier League or Serie A. In the pooled sample ($n = 790$), this same weak relationship ($r = 0.075$, under 1% shared variance) reaches conventional significance largely because of the larger sample size. This points to the fact that the effect size, not the p-value, is the figure worth leading with here.

3.6 Top-ranked players and rank divergence

As established in Section 2.3, rank divergence between Elo and success rate is only meaningful among players who share a duel network, and is therefore reported separately for each league rather than as a single combined ranking. A player's position within a simple concatenation of all three leagues' rows would not reflect a genuine competitive comparison, since no player from one league ever faced a player from another. The following is a report of the largest divergences in terms of rank within each league from Elo to success rate. These divergences are presented as (Elo rank vs. Success rate rank, difference between ranks).

Premier League

Top 10 attacking divergences (of 297 qualifying): Jamie Vardy (29 vs. 227, diff. 198); Wilfried Zaha (2 vs. 157, diff. 155); Anthony Martial (63 vs. 209, diff. 146); James McClean (79 vs. 221, diff. 142); Stéphane Sessègnon (27 vs. 152, diff. 125); Salomón Rondón (127 vs. 250, diff. 123); Junior Stanislas (66 vs. 189, diff. 123); Jordan Ayew (86 vs. 207, diff. 121); James Milner (88 vs. 206, diff. 118); Harry Kane (68 vs. 185, diff. 117). The clearest case running the other way is Nicolás Otamendi (172 vs. 63, diff. 109).

Top 10 defensive divergences (of 336 qualifying): Adam Smith (275 vs. 105, diff. 170); Yohan Cabaye (44 vs. 174, diff. 130); Ashley Westwood (258 vs. 139, diff. 119); James McArthur (140 vs. 257, diff. 117); Lucas Leiva (321 vs. 204, diff. 117); Yann M'Vila (310 vs. 194, diff. 116); Theo Walcott (205 vs. 313, diff. 108); Nemanja Matić (173 vs. 65, diff. 108); André Ayew (308 vs. 204, diff. 104); Roberto Firmino (85 vs. 187, diff. 102).

La Liga

Top 10 attacking divergences (of 316 qualifying): Fabián Orellana (56 vs. 185, diff. 129); Rubén Castro (135 vs. 260, diff. 125); Filipe Luís (35 vs. 153, diff. 118); Alen Halilović (74 vs. 184, diff. 110); Iago Aspas (111 vs. 221, diff. 110); Antonio Luna (159 vs. 51, diff. 108); José Fonte (120 vs. 14, diff. 106); Yannick Carrasco (256 vs. 152, diff. 104); Gabriel Fernández (6 vs. 103, diff. 97); Nicolás López (165 vs. 69, diff. 96).

Top 10 defensive divergences (of 355 qualifying): Pedro Mosquera (61 vs. 217, diff. 156); Sergio Gontán (309 vs. 172, diff. 137); Daniel Ceballos (353 vs. 225, diff. 128); Juan Antonio Rodríguez (328 vs. 222, diff. 106); Josep Señé (218 vs. 320, diff. 102); Javier Fuego (37 vs. 137, diff. 100); Miguel Torres (245 vs. 148, diff. 97); Nordin Amrabat (224 vs. 320, diff. 96); Roberto Trashorras (340 vs. 244, diff. 96); Beñat Etxebarria (283 vs. 187, diff. 96).

Serie A

Top 10 attacking divergences (of 287 qualifying): Lorenzo Insigne (19 vs. 145, diff. 126); Antonio Di Gaudio (60 vs. 181, diff. 121); Federico Bernardeschi (226 vs. 112, diff. 114); Bruno Fernandes (54 vs. 163, diff. 109); Sebastian De Maio (141 vs. 40, diff. 101); Mohamed Salah (108 vs. 208, diff. 100); Senad Lulić (272 vs. 174, diff. 98); Adem Ljajić (63 vs. 158, diff. 95); Edin Džeko (80 vs. 173, diff. 93); Leandro Paredes (133 vs. 41, diff. 92).

Top 10 defensive divergences (of 336 qualifying): Leandro Paredes (320 vs. 205, diff. 115); Borja Valero (277 vs. 175, diff. 102); Davide Astori (62 vs. 156, diff. 94); Saphir Taïder (174 vs. 268, diff. 94); Danilo Soddimo (238 vs. 145, diff. 93); Marten de Roon (308 vs. 215, diff. 93); Rafael Tolói (46 vs. 137, diff. 91); Nicolò Cherubin (188 vs. 274, diff. 86); Wesley Hoedt (219 vs. 134, diff. 85); Juan Cuadrado (294 vs. 210, diff. 84).

Across all three leagues, wingers and strikers dominate the attacking-divergence lists, while defensive and central midfielders dominate the defensive-divergence lists. Sections 4.2 and 4.3 return to this pattern.

3.7 Position-group summary (pooled)

Table 9 reports an aggregate description of the positional pattern across all three leagues; it is not a player ranking.

Table 9: Pooled position-group summary (aggregate description of positional pattern across all three leagues.

Group	**n**	**Att. qual.**	**Def. qual.**	**Att. Elo**	**Def. Elo**	**Att. SR**	**Def. SR**
Centre backs	256	59	191	1032.10 (34.47)	1027.82 (50.88)	0.790 (0.210)	0.555 (0.168)
Full backs	260	166	212	1009.05 (50.92)	1016.49 (56.28)	0.570 (0.193)	0.493 (0.161)
Defensive mids	193	134	160	1048.98 (53.16)	959.07 (58.98)	0.690 (0.181)	0.383 (0.153)
Centre mids	123	91	99	1028.47 (52.24)	951.20 (58.64)	0.600 (0.179)	0.360 (0.146)
Attacking mids	81	55	52	1020.32 (54.54)	938.75 (65.23)	0.593 (0.199)	0.302 (0.211)
Wingers	279	228	197	1022.75 (60.75)	942.65 (56.82)	0.568 (0.160)	0.319 (0.169)
Strikers	203	145	90	1012.70 (53.90)	951.11 (52.88)	0.548 (0.187)	0.269 (0.235)

This table describes an average positional pattern across three independent leagues. The position-group qualifying counts sum to 878 attackers and 1,001 defenders, which falls slightly short of the pooled qualifying totals of 899 and 1,026: the gap is made up by goalkeepers as they were included in the overall Elo rankings but not in positional analysis. Results that can be drawn from this are that Centre backs post the highest defensive success rate and Elo, while attacking mids record the lowest defensive Elo and strikers the lowest defensive success rate.

3.8 Rank movement by position group

Table 10 tests whether the player-level divergence from Section 3.6 follows a positional pattern. Rank movement is defined as $SRrank - Elorank$, which is then averaged by position group within each league before pooling. A positive value means that success rate underrates a group relative to Elo while a negative value means it overrates one.

Table 10: Mean rank movement (SR rank − Elo rank) by position group and role, pooled across leagues

Role	Position group	n	Mean movement	Median	SD
Attack	Strikers	145	+54.52	36.0	122.53
Attack	Wingers	228	+39.42	24.5	147.01
Attack	Attacking mids	55	+28.87	38.0	129.58
Attack	Wing backs	21	+3.90	38.0	124.49
Attack	Centre mids	91	−9.19	0.0	146.72
Attack	Full backs	166	−22.61	−20.5	112.11
Attack	Defensive mids	134	−73.43	−60.5	102.41
Attack	Centre backs	59	−115.47	−112.0	93.85
Defense	Strikers	90	+66.12	59.5	87.99
Defense	Attacking mids	52	+12.35	−4.5	126.77
Defense	Wingers	197	+11.65	8.0	116.59
Defense	Centre mids	99	−6.00	−23.0	126.45
Defense	Full backs	212	−15.75	−10.5	107.00
Defense	Defensive mids	160	−18.93	−26.0	149.37
Defense	Centre backs	191	−23.42	−23.0	85.98
Defense	Wing backs	25	−40.56	−40.0	114.52

4 Discussion

4.1 Does Elo diverge from success rate?

The pooled correlation figures from Section 3.3 translate into $r^2 \approx 0.711$ for attacking Elo and $r^2 \approx 0.808$ for defensive Elo, meaning that roughly 19–28% of the variance in one metric goes unexplained by the other, a gap consistent with each individual league's own r^2. Read alongside McDermott et al. (2021), this result proves the broader claim that opposition quality matters for 1v1 outcomes, independent of any single player's results.

4.2 Who benefits from Elo re-ranking?

Table 10 reveals a consistent trend in the comparison between Elo and success rate in terms of positional rating differences. For Attacking and Defensive ratings success rate tends to overrate more defensive positions (Centre backs, Defensive mids, etc.) and penalize attacking positions (Strikers, Wingers, etc.) relative to Elo. From prior examination in Section 3.6 we recognize that these trends are also followed for almost all of our players with large rating gaps. For example. Jamie Vardy and Wilfried Zaha, both attacking players, drastically jump up the rankings when

switching from SR to Elo while Nicolás Otamendi, a Centre Back, goes down in rank when switching to Elo.

This gives the observed divergence in Section 3.6 structural backing. Success rate appears to conflate ability with opposition difficulty in a position-dependent way, not just randomly, with the direction of the bias being inverse between forward and back-line roles. Spread within groups is large (full detail in `position_rank_movement.csv`), so this is a group-level tendency rather than a rule that holds for every individual player.

4.3 Positional differences

The pooled effect-size comparison from Section 3.4 shows attacking Elo's effect size shrinking under pooling ($\eta^2 = 0.093$, against a range of 0.096–0.133 across the three individual leagues), while defensive Elo's effect size stays essentially unchanged ($\eta^2 = 0.393$ pooled, against 0.382–0.430 individually). One possible explanation for this is that the quality associated with a given attacking position, a winger's or striker's role, for example, varies more from league to league in tactical style than the quality associated with a given defensive position does. If this is so, pooling would add extra variance for attackers that a single-league ANOVA never encounters. Positional patterns are reported in the pooled position-group table (Section 3.7) and again in the Mean rank movement table (Section 4.2) but further research would be required to prove the truth of the prior explanations.

4.4 Model parameter sensitivity

As with prior applications of this kind of model, the choice of K-factor and minimum-duel threshold is a researcher decision rather than something derived from the data itself (Bisberg & Cardona-Rivera, 2019). Because the three leagues differ in duel frequency and defensive baseline (Section 3.1), a single global K-factor and duel threshold applied across all three may not suit every league equally well. Tuning these parameters separately for each league is left as an unexplored option for future work.

4.5 Limitations

StatsBomb's event data identifies the defender in a duel as the nearest opposing player at the point of the attempt, which can occasionally misattribute the primary defender during congested phases of play. The dataset also covers only the 2015/16 season, so it cannot speak to whether the patterns reported here hold from one season to the next. The Elo model itself assumes that each duel is independent of previous duels between the same two players, an assumption that may not prove true given further research, since certain attacker-defender pairings can occur several times within a match or across a season, with dynamics and approaches to duels changing.

One further limitation is central to how this study is scoped. Because leagues never share duels, any numeric comparison of Elo ratings between players in different leagues is unvalidated by this study's design. Multiple of these limitations can be addressed with access to more data, as constructing these rankings over multiple season and cross league competitions (Champions League, World Cup, etc.) would allow for rankings to be numerically scaled and comparable across leagues.

5 Practical Applications and Recommendations

For scouting within a single league, Elo enables a fairer comparison of players who face systematically weaker or stronger opposition than raw success rate allows, as illustrated by the per-league examples in Section 4.2. For this study, extending Elo to cross-league comparison is better treated as future work than as a current capability (given current lack of data).

For opposition analysis, defensive Elo can identify players who consistently perform highly against elite attackers within their own league even where a raw succes rate may looks unremarkable. Cabaye and Matić in the Premier League are examples from the results above. For metric development more broadly, Elo is best positioned as a complement to existing volume and success-rate metrics within a league rather than a replacement for them, given the strong but imperfect correlations reported in Section 3.3.

On data requirements, Elo needs only duel outcomes to compute, which makes it straightforward to extend the same within-league approach to further leagues. The same design principle, building within-league ratings first and treating cross-league comparison as a separate, harder problem, extends naturally to other inverse-metric pairs in football analytics. For example pressing versus ball-carrying, aerial duels (Kim & Kim, 2025), and shooting versus goalkeeping. Separate studies would need to be performed for each to explore if they benefit in similar ways.

6 Conclusion

This study set out to construct within-league Elo ratings across three leagues and to test whether the resulting Elo-success-rate divergence pattern is statistically robust in aggregate. Both parts of this aim were achieved. Within every league studied, Elo rankings diverge meaningfully from success-rate rankings for specific, identifiable players, by as much as 198 places in the Premier League and 156 in La Liga. The weak and inconsistent relationship between attacking and defensive Elo supports treating 1v1 attacking and defending as genuinely separate abilities rather than two expressions of the same skill. This difference thus backs the decision to treat them as two seperate Elo ratings within this study.

The pooled statistics confirm that both the Elo to success-rate divergence pattern and the positional-effect pattern recur consistently across the three independent leagues which strengthens confidence that the results can be generalized to all of football rather than just one specific competition. Cross-league differences in defensive baseline, weakest in the Premier League and strongest in Seria A, are further evidence that opposition-quality adjustment matters across leagues as well as within them, though genuine cross-league player comparison remains a direction for future work requiring further input data.

Taken together, these findings suggest that opposition-adjusted rating models deserve a more central place alongside raw performance indicators in football analytics practice, within any given competitive context.

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Appendices

Appendix A: Position-Group Mapping

Table 11 reports the mapping used to collapse StatsBomb's granular position labels into the eight position groups used throughout this study (Section 2.4). A player's position group was assigned from the modal value of their per-duel `position` field across all duel events in which they participated, mapped through this table. Goalkeepers were retained in the overall Elo rankings but excluded from all position-group analyses, since a goalkeeper's involvement in an open-play 1v1 duel is not comparable to an outfield player's.

Table 11: Mapping from StatsBomb granular position labels to the eight position groups used in this study.

StatsBomb label	**Position group**
Centre backs	
Left Centre Back	Centre backs
Right Centre Back	Centre backs
Centre Back	Centre backs
Full backs	
Left Back	Full backs
Right Back	Full backs
Wing backs	
Left Wing Back	Wing backs
Right Wing Back	Wing backs
Defensive mids	
Left Defensive Midfield	Defensive mids
Centre Defensive Midfield	Defensive mids
Right Defensive Midfield	Defensive mids

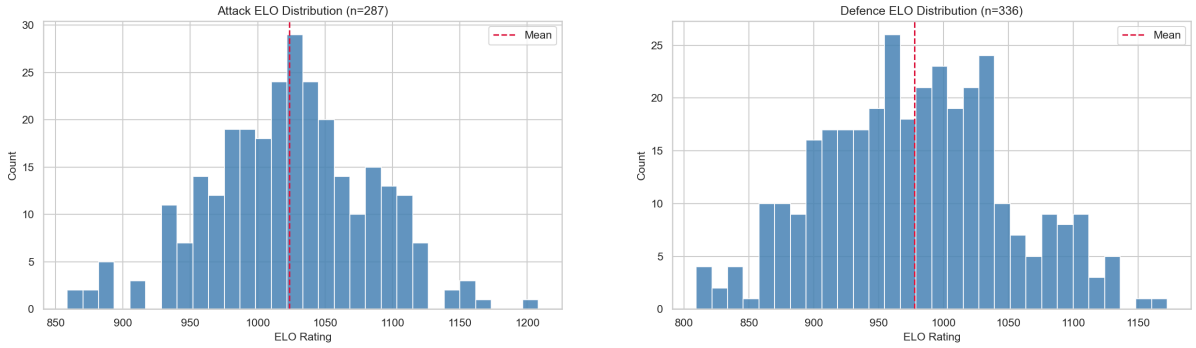
StatsBomb label	Position group
Centre mids	
Left Centre Midfield	Centre mids
Centre Midfield	Centre mids
Right Centre Midfield	Centre mids
Attacking mids	
Centre Attacking Midfield	Attacking mids
Left Attacking Midfield	Attacking mids
Right Attacking Midfield	Attacking mids
Wingers	
Left Wing	Wingers
Right Wing	Wingers
Left Midfield	Wingers
Right Midfield	Wingers
Strikers	
Centre Forward	Strikers
Left Centre Forward	Strikers
Right Centre Forward	Strikers
Excluded from grouped analysis	
Goalkeeper	— (excluded)

[NOTE: Left Midfield and Right Midfield are grouped with Wingers rather than Centre mids, and Left/Right Centre Forward are grouped with Strikers rather than Wingers, per the comments in the source mapping. If you'd rather these ambiguous labels sat in the other group, this is the place to change it, since Table 11 is generated directly from the same dictionary used in `positions.py`*.]*

Appendix B: Python Pipeline Summary

The full analysis pipeline, including duel-pairing extraction, Elo computation, and position-group mapping, is publicly available (Dormody, 2026).

Appendix C: Supplementary Tables and Figures



(a) Attacking Elo.

(b) Defensive Elo.

Figure 5: Serie A Elo rating distributions.

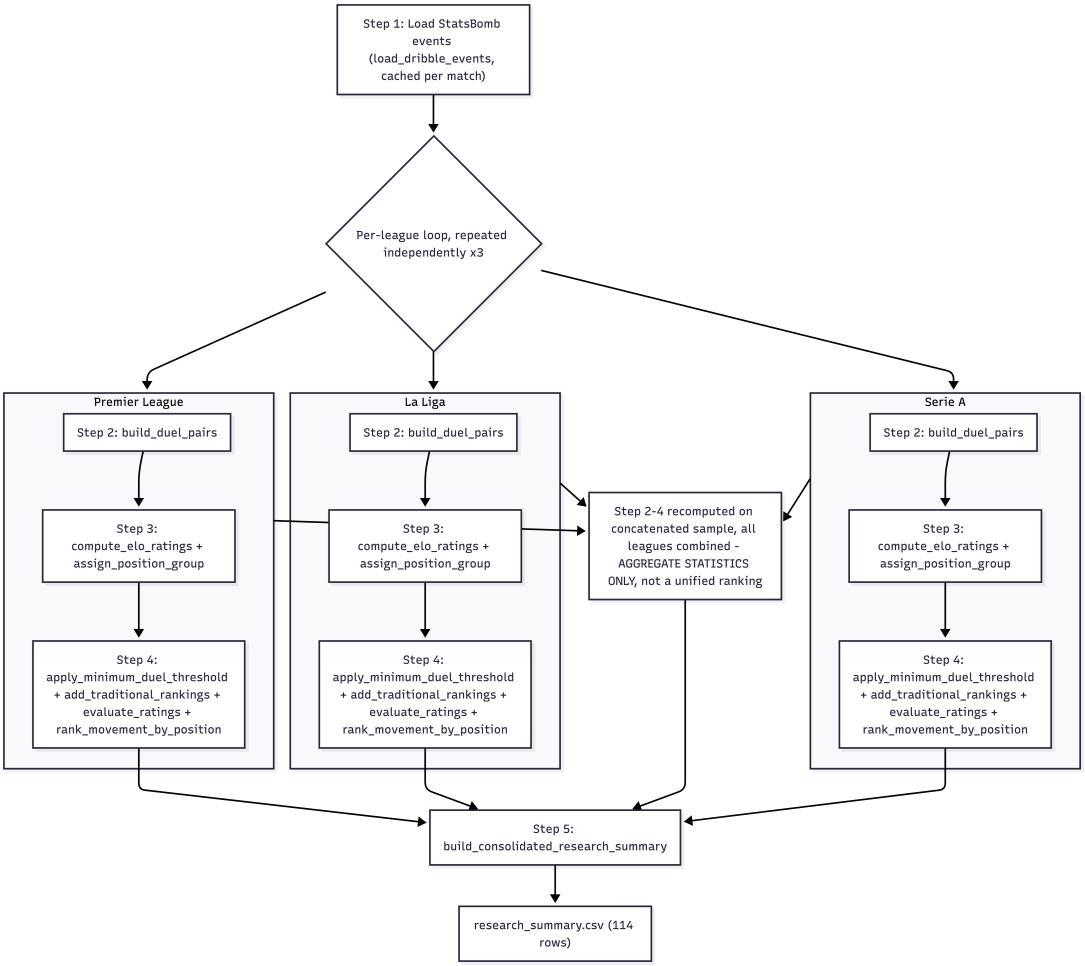
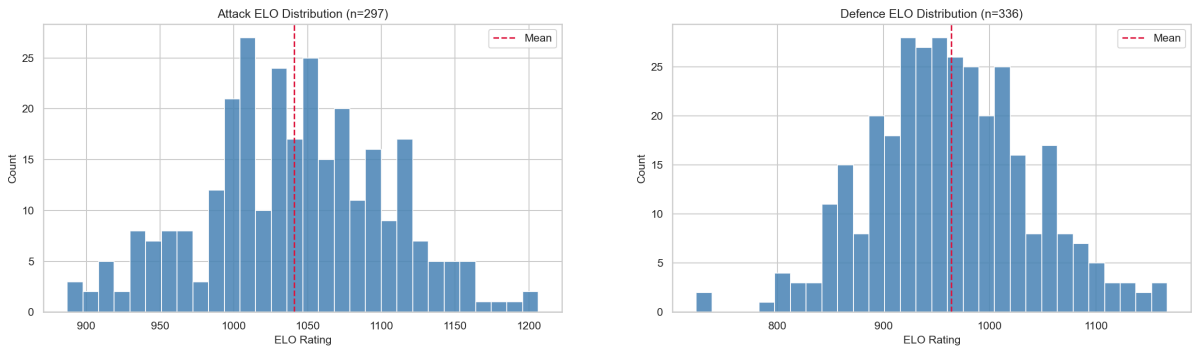


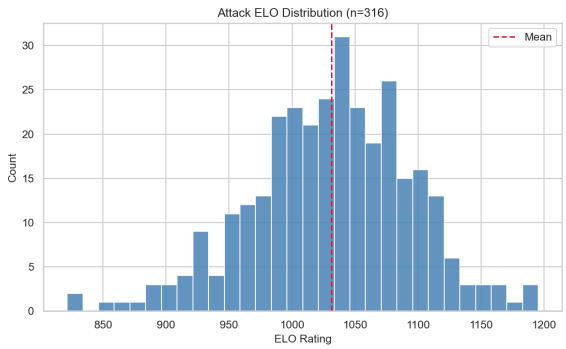
Figure 2: Analysis pipeline: per-league loop (Steps 2–4, independent per league) feeds a separate concatenated pooled run (aggregate statistics only), converging on a consolidated `research_summary.csv`.



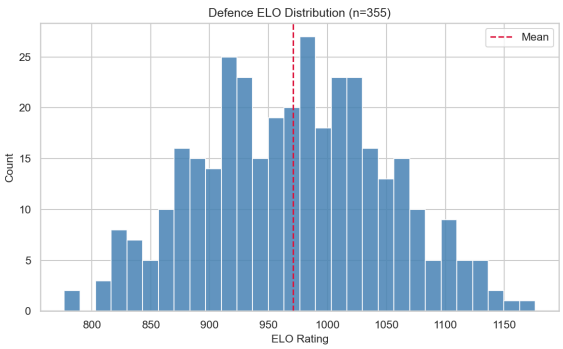
(a) Attacking Elo.

(b) Defensive Elo.

Figure 3: Premier League Elo rating distributions.



(a) Attacking Elo.



(b) Defensive Elo.

Figure 4: La Liga Elo rating distributions.

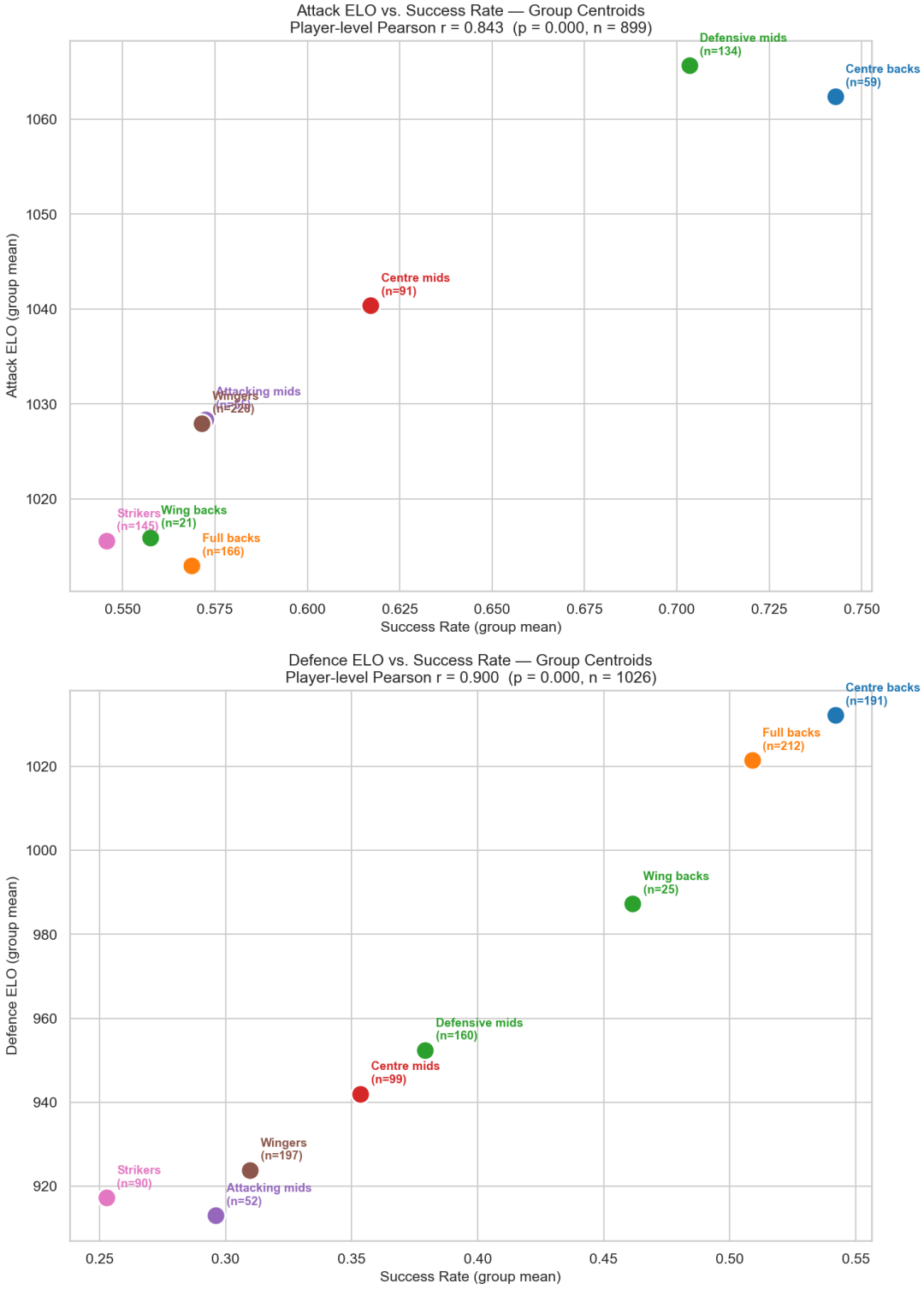


Figure 6: Position-group centroids for Elo against success rate, pooled across the three leagues. Player-level Pearson correlations (Table 5) are annotated on each panel. Underlying player-level data by league are reported in Tables 3–4.